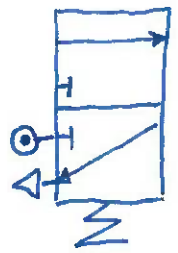
Pneumatics

COMPONENTS

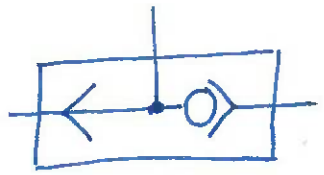
3/2 VALVE



5/2 VALVE

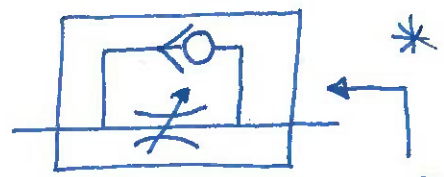


SHUTTLE VALVE



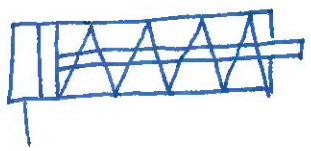
* CREATES OR CONTROL *

UNI-DIRECTIONAL RESTRICTOR



USED TOGETHER TO CREATE TIME DELAY

SINGLE ACTING CYLINDER



DOUBLE ACTING CYLINDER



RESTRICTOR



* SLOWS AIR FLOW *

RESERVOIR



ACTUATORS

PUSH BUTTON



SOLENOID



ROLLER TRIP



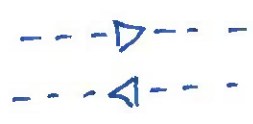
PLUNGER



LEVER



DIAPHRAGM



ROLLER



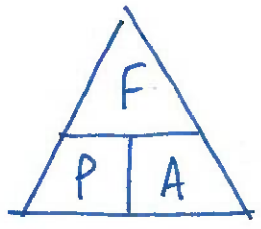
PILOT AIR



FOOT PEDAL



CALCULATIONS



FORCE - N

PRESSURE - N/mm^2

AREA - mm^2

$AREA = \pi r^2$

$r = \sqrt{\frac{A}{\pi}}$

INSTROKE CALCULATION

EFFECTIVE AREA = PISTON AREA − ROD AREA.

DESCRIBING A CIRCUIT

- INPUT → PROCESS → OUTPUT
- USE VALVE NAMES GIVEN IN QUESTION.
- MARK UP DIAGRAM TO CHECK YOU MENTION ALL COMPONENTS.
- WORD BANK:

'ACTUATE' 'CHANGE STATE'

'INSTROKE' 'OUTSTROKE'

'TIME DELAY'

'SLOWLY'